

Midterm Exam 2

ECON 320 – Introduction to Mathematical Economics

November 05, 2025

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Instructions:

- You are allowed **75 minutes** to complete the exam.
 - Show all work. Unsupported answers receive little credit.
 - Unless stated otherwise, all variables are real and differentiable as needed.
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Problem 1 (12 points)

A firm's profit function is

$$\pi(Q) = 120Q - 6Q^2 - 180.$$

- (a) Derive the first-order condition and find the critical point Q^* . (4 pts)
- (b) Compute the second derivative and determine whether Q^* is a maximum or minimum. (3 pts)
- (c) Find the maximum profit π^* . (3 pts)
- (d) Briefly interpret Q^* and the curvature of $\pi(Q)$ economically. (2 pts)

Problem 2 (12 points)

A population grows according to

$$P(t) = 200e^{0.03t},$$

and a technology index grows as

$$A(t) = e^{0.02t}.$$

Output is given by a Cobb–Douglas form $Y(t) = A(t)P(t)^{0.5}$.

- (a) Find the instantaneous growth rate of $Y(t)$ by differentiating $\ln Y(t)$. (4 pts)
- (b) Compute $Y(10)$ (rounded to two decimals). (3 pts)
- (c) Suppose the population growth rate rises from 3% to 4%, while technology remains the same. Compute the new growth rate of $Y(t)$. (3 pts)
- (d) Interpret the difference in growth rates: what does it imply about the contribution of population versus technology? (2 pts)

Problem 3 (13 points)

Let

$$f(x, y) = 12x + 8y - x^2 - y^2 - xy.$$

- (a) Compute f_x and f_y and find all stationary points. (4 pts)
- (b) Form the Hessian matrix H and its determinant. (4 pts)
- (c) Use the Hessian to classify the stationary point (maximum, minimum, or saddle). (3 pts)
- (d) Provide an economic interpretation of the curvature of $f(x, y)$. (2 pts)

Problem 4 (13 points)

A consumer maximizes

$$U(x, y) = x^{0.5}y^{0.5}$$

subject to the budget constraint

$$2x + y = 100.$$

- (a) Form the Lagrangian and derive the first-order conditions. (4 pts)
- (b) Solve for x^*, y^*, λ^* . (5 pts)
- (c) Verify (briefly) that the bordered Hessian condition confirms a maximum. (2 pts)
- (d) Interpret the meaning of λ^* and use it to estimate the change in utility if income increases by 5. (2 pts)